

where $X, X' \stackrel{\text{iid}}{\sim} p$ and $Y, Y' \stackrel{\text{iid}}{\sim} q$ are all independent of each other. We deduce that

$$\begin{aligned}
 \mathbb{E}_{p \times q} \left[\frac{1}{m^2(m-1)^2} \sum_{1 \leq i \neq j \leq m+n} k_\lambda(U_i, U_j)^2 \right] &\leq \frac{(m+n)(m+n-1)}{m^2(m-1)^2} \frac{M\kappa_2}{\lambda_1 \cdots \lambda_d} \\
 &\leq \frac{4M\kappa_2}{\lambda_1 \cdots \lambda_d} \frac{(m+n)^2}{m^4} \\
 &= \frac{64M\kappa_2}{\lambda_1 \cdots \lambda_d} \frac{(m+n)^2}{(2m)^4} \\
 &\leq \frac{64M\kappa_2}{\lambda_1 \cdots \lambda_d} \frac{(m+n)^2}{(m+C^{-1}n)^4} \\
 &\leq \frac{64M\kappa_2 C^4}{\lambda_1 \cdots \lambda_d} \frac{1}{(m+n)^2}
 \end{aligned}$$

where we use the fact that $n \leq Cm$. Using Markov's inequality, we get that, for any $\delta \in (0, 1)$, we have

$$\begin{aligned}
 &1 - \frac{\delta}{2} \\
 &\leq \mathbb{P}_{p \times q} \left(\frac{1}{m^2(m-1)^2} \sum_{1 \leq i \neq j \leq m+n} k_\lambda(U_i, U_j)^2 \leq \frac{2}{\delta} \mathbb{E}_{p \times q} \left[\frac{1}{m^2(m-1)^2} \sum_{1 \leq i \neq j \leq m+n} k_\lambda(U_i, U_j)^2 \right] \right) \\
 &\leq \mathbb{P}_{p \times q} \left(\frac{1}{m^2(m-1)^2} \sum_{1 \leq i \neq j \leq m+n} k_\lambda(U_i, U_j)^2 \leq \frac{2}{\delta} \frac{64M\kappa_2 C^4}{(m+n)^2 \lambda_1 \cdots \lambda_d} \right) \\
 &\leq \mathbb{P}_{p \times q} \left(\hat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} 8c_1 C^2 \sqrt{2M\kappa_2} \frac{\ln(\frac{2}{\alpha})}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \\
 &\leq \mathbb{P}_{p \times q} \left(\hat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} 16c_1 C^2 \sqrt{2M\kappa_2} \frac{\ln(\frac{1}{\alpha})}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right)
 \end{aligned}$$

as $\ln(\frac{2}{\alpha}) \leq 2\ln(\frac{1}{\alpha})$ since $\alpha \in (0, 0.5)$. We now let $C_2(M, d) := 16c_1 C^2 \sqrt{2M\kappa_2}$. Then, for all $B \in \mathbb{N}$ such that $B \geq \frac{3}{\alpha^2} (\ln(\frac{4}{\delta}) + \alpha(1-\alpha))$, we have

$$\begin{aligned}
 &\mathbb{P}_{p \times q \times r} \left(\hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} C_2(M, d) \frac{\ln(\frac{1}{\alpha})}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \\
 &\geq \mathbb{P}_{p \times q \times r} \left(\left\{ \hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \right\} \right. \\
 &\quad \left. \cap \left\{ \hat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_m, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} C_2(M, d) \frac{\ln(\frac{1}{\alpha})}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right\} \right) \\
 &\geq 1 - \frac{\delta}{2} - \frac{\delta}{2} \\
 &= 1 - \delta
 \end{aligned}$$

where we use the standard fact that for events $\mathcal{B}$ and $\mathcal{C}$ satisfying $\mathbb{P}(\mathcal{B}) \geq 1 - \delta_1$ and $\mathbb{P}(\mathcal{C}) \geq 1 - \delta_2$, we have $\mathbb{P}(\mathcal{B} \cap \mathcal{C}) = 1 - \mathbb{P}(\mathcal{B}^c \cup \mathcal{C}^c) \geq 1 - \mathbb{P}(\mathcal{B}^c) - \mathbb{P}(\mathcal{C}^c) \geq 1 - \delta_1 - \delta_2$.

Quantile bound for MMD estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ defined in Equation (6).

For this case, in order to upper bound $\widehat{q}_{1-\alpha}^{\lambda, \infty}(\mathbb{X}_n, \mathbb{Y}_n)$, we can use the result of de la Peña and Giné (1999, Corollary 3.2.6) and Markov's inequality as done by Fromont et al. (2012, Appendix D). We obtain that there exists a positive constant $\tilde{c}_1$ such that

$$\mathbb{P}_r \left(\left| \sum_{1 \leq i \neq j \leq n} \epsilon_i \epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j) \right| \geq \tilde{c}_1 \sqrt{\sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)^2} \ln \left(\frac{2}{\alpha} \right) \middle| \mathbb{X}_n, \mathbb{Y}_n \right) \leq \alpha$$

where $\epsilon_1, \dots, \epsilon_n$ are i.i.d. Rademacher variables, and so $\sum_{1 \leq i \neq j \leq n} \epsilon_i \epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j)$ is a Rademacher chaos. We deduce that

$$\widehat{q}_{1-\alpha}^{\lambda, \infty}(\mathbb{X}_n, \mathbb{Y}_n) \leq \tilde{c}_1 \sqrt{\frac{1}{n^2(n-1)^2} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)^2} \ln \left(\frac{2}{\alpha} \right)$$

almost surely. Using Equation (23), we have

$$\begin{aligned} & \mathbb{E}_{p \times q} \left[\frac{1}{n^2(n-1)^2} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)^2 \right] \\ &= \frac{1}{n^2(n-1)^2} \mathbb{E}_{p \times q} \left[\sum_{1 \leq i \neq j \leq n} (k(X_i, X_j) + k(Y_i, Y_j) - k(X_i, Y_j) - k(X_j, Y_i))^2 \right] \\ &\leq \frac{4}{n(n-1)} \left(\mathbb{E}_{X, X'} [k_\lambda(X, X')^2] + \mathbb{E}_{Y, Y'} [k_\lambda(Y, Y')^2] + 2\mathbb{E}_{X, Y} [k_\lambda(X, Y)^2] \right) \\ &\leq \frac{32M\kappa_2}{n^2\lambda_1 \cdots \lambda_d} \end{aligned}$$

where $X, X' \stackrel{\text{iid}}{\sim} p$ and $Y, Y' \stackrel{\text{iid}}{\sim} q$ are all independent of each other, and where κ_2 is the constant defined in Equation (21) depending on d . Similarly to the previous case, we can then use Markov's inequality to get that, for any $\delta \in (0, 1)$, we have

$$\begin{aligned} & 1 - \frac{\delta}{2} \\ &\leq \mathbb{P}_{p \times q} \left(\frac{1}{n^2(n-1)^2} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)^2 \leq \frac{2}{\delta} \mathbb{E}_{p \times q} \left[\frac{1}{n^2(n-1)^2} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)^2 \right] \right) \\ &\leq \mathbb{P}_{p \times q} \left(\frac{1}{n^2(n-1)^2} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)^2 \leq \frac{2}{\delta} \frac{32M\kappa_2}{n^2\lambda_1 \cdots \lambda_d} \right) \\ &\leq \mathbb{P}_{p \times q} \left(\widehat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_n, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} 8\tilde{c}_1 \sqrt{M\kappa_2} \frac{\ln(\frac{4}{\alpha})}{n\sqrt{\lambda_1 \cdots \lambda_d}} \right) \\ &\leq \mathbb{P}_{p \times q} \left(\widehat{q}_{1-\alpha/2}^{\lambda, \infty}(\mathbb{X}_n, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} 48\tilde{c}_1 \sqrt{M\kappa_2} \frac{\ln(\frac{1}{\alpha})}{2n\sqrt{\lambda_1 \cdots \lambda_d}} \right) \end{aligned}$$

as $\ln(\frac{4}{\alpha}) \leq 3\ln(\frac{1}{\alpha})$ since $\alpha \in (0, 0.5)$. Letting $\tilde{C}_2(M, d) := 48\tilde{c}_1\sqrt{M\kappa_2}$ and applying the same reasoning as earlier, we get

$$\mathbb{P}_{p \times q \times r} \left(\hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \leq \frac{1}{\sqrt{\delta}} \tilde{C}_2(M, d) \frac{\ln(\frac{1}{\alpha})}{2n\sqrt{\lambda_1 \cdots \lambda_d}} \right) \leq 1 - \delta$$

for all $B \in \mathbb{N}$ satisfying $B \geq \frac{3}{\alpha^2} (\ln(\frac{4}{\delta}) + \alpha(1 - \alpha))$.

E.5 Proof of Theorem 5

First, as shown by Gretton et al. (2012a, Lemma 6), the Maximum Mean Discrepancy can be written as

$$\begin{aligned} \text{MMD}_\lambda^2(p, q) &= \mathbb{E}_{X, X'} [k_\lambda(X, X')] - 2 \mathbb{E}_{X, Y} [k_\lambda(X, Y)] + \mathbb{E}_{Y, Y'} [k_\lambda(Y, Y')] \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} k_\lambda(x, x') p(x) p(x') \, dx dx' \\ &\quad - 2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} k_\lambda(x, y) p(x) q(y) \, dx dy \\ &\quad + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} k_\lambda(y, y') q(y) q(y') \, dy dy' \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} k_\lambda(u, u') (p(u) - q(u)) (p(u') - q(u')) \, du du' \end{aligned}$$

for $X, X' \stackrel{\text{iid}}{\sim} p$ and $Y, Y' \stackrel{\text{iid}}{\sim} q$ all independent of each other. Using the function φ_λ defined in Equation (9) and $\psi := p - q$, we obtain

$$\begin{aligned} \text{MMD}_\lambda^2(p, q) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \varphi_\lambda(u - u') \psi(u) \psi(u') \, du du' \\ &= \int_{\mathbb{R}^d} \psi(u) \int_{\mathbb{R}^d} \psi(u') \varphi_\lambda(u - u') \, du' du \\ &= \int_{\mathbb{R}^d} \psi(u) (\psi * \varphi_\lambda)(u) \, du \\ &= \langle \psi, \psi * \varphi_\lambda \rangle_2 \\ &= \frac{1}{2} \left(\|\psi\|_2^2 + \|\psi * \varphi_\lambda\|_2^2 - \|\psi - \psi * \varphi_\lambda\|_2^2 \right) \end{aligned}$$

where the last equality is obtained by expanding $\|\psi - \psi * \varphi_\lambda\|_2^2$. By Lemma 2, a sufficient condition to ensure that $\mathbb{P}_{p \times q \times r}(\Delta_\alpha^{\lambda, B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 0) \leq \beta$ is

$$\mathbb{P}_{p \times q \times r} \left(\text{MMD}_\lambda^2(p, q) \geq \sqrt{\frac{2}{\beta} \text{var}_{p \times q}(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n))} + \hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \right) \geq 1 - \frac{\beta}{2}$$

and an equivalent sufficient condition is

$$\mathbb{P}_{p \times q \times r} \left(\|\psi\|_2^2 \geq \|\psi - \psi * \varphi_\lambda\|_2^2 - \|\psi * \varphi_\lambda\|_2^2 + 2\sqrt{\frac{2}{\beta} \text{var}_{p \times q}(\widehat{\text{MMD}}_\lambda^2)} + 2\hat{q}_{1-\alpha}^{\lambda, B} \right) \geq 1 - \frac{\beta}{2}.$$